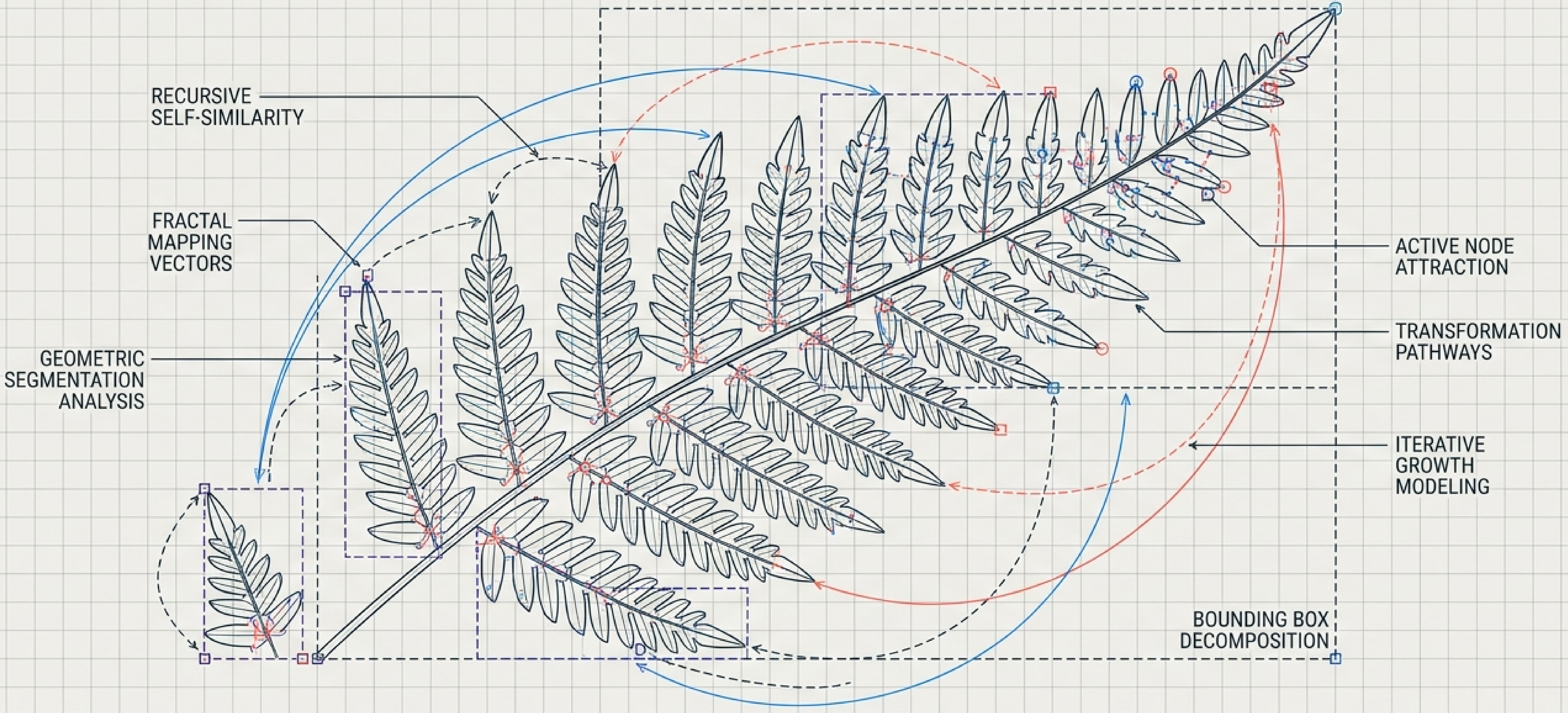
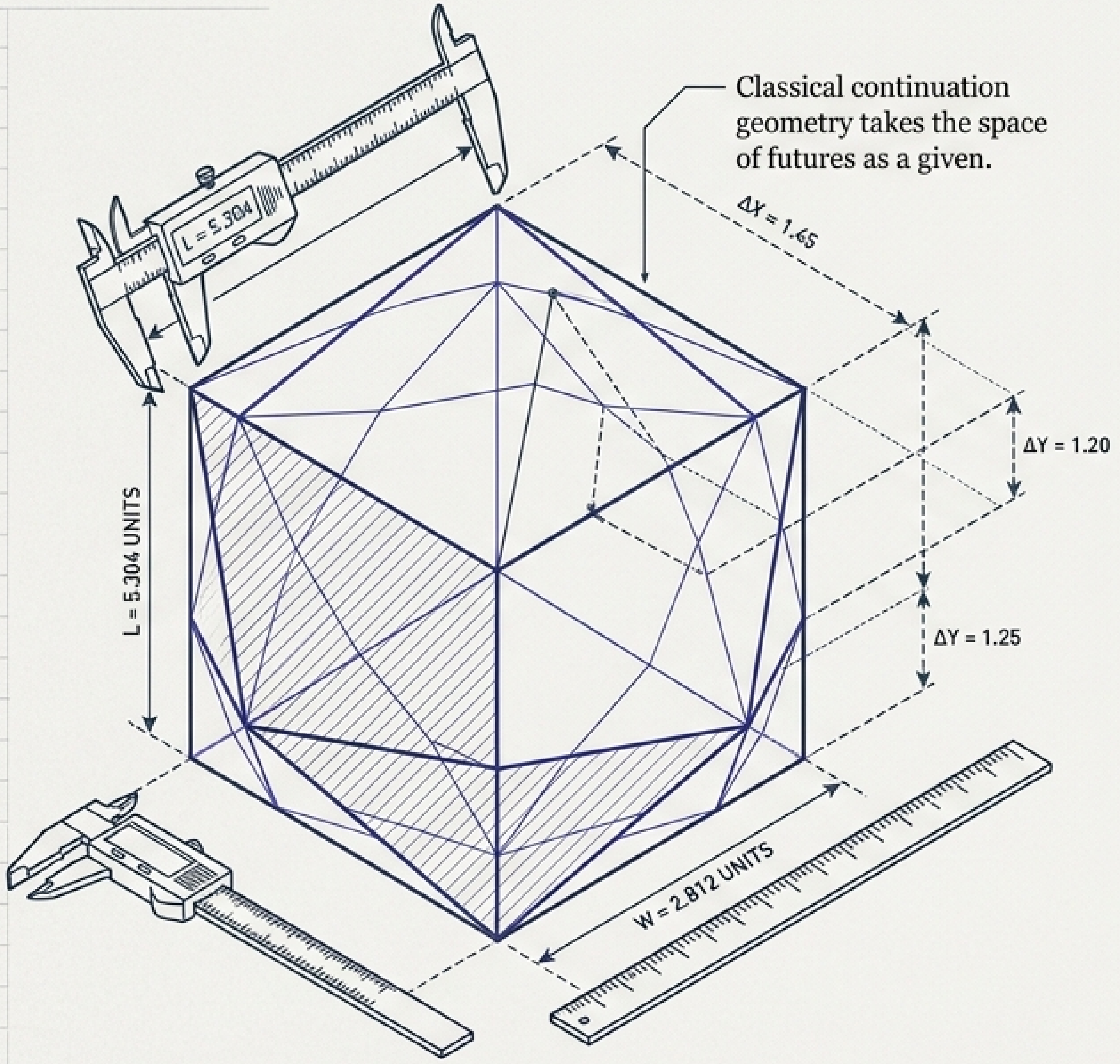
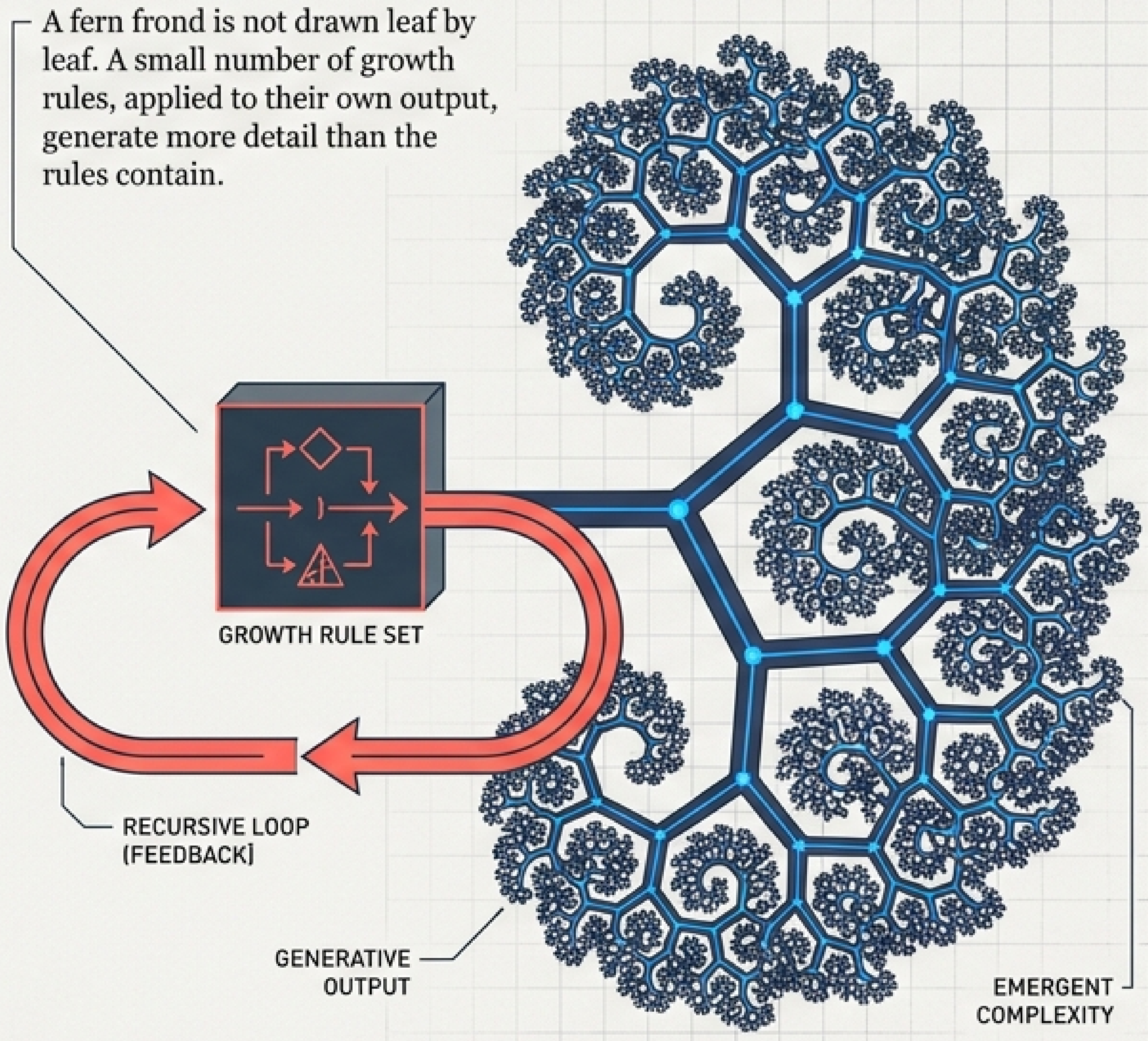
GENERATIVE CONTINUATION GEOMETRY

WORKING DRAFT SYNTHESIS



A space of futures can be grown, not just measured.



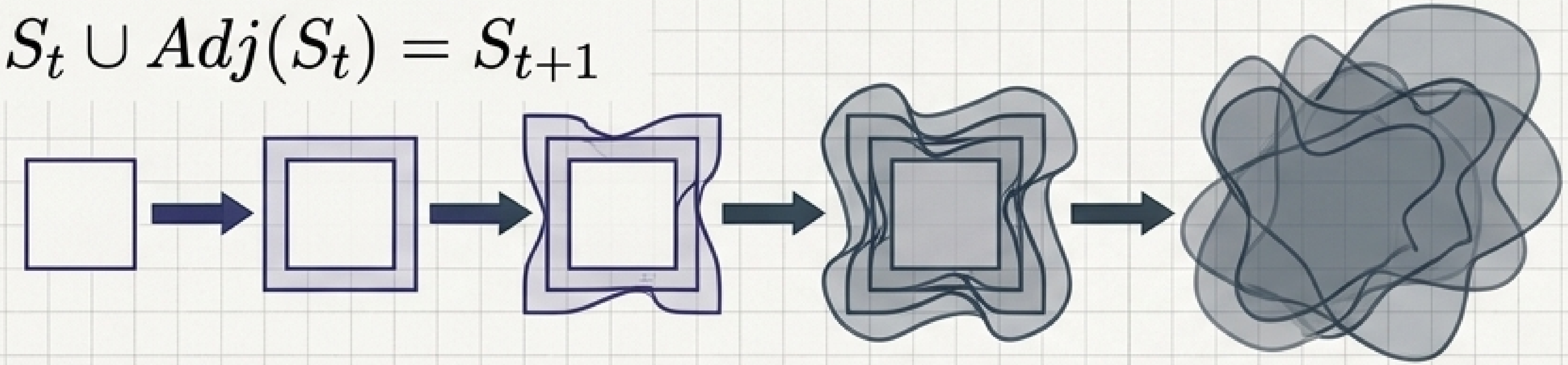


NotebookLM

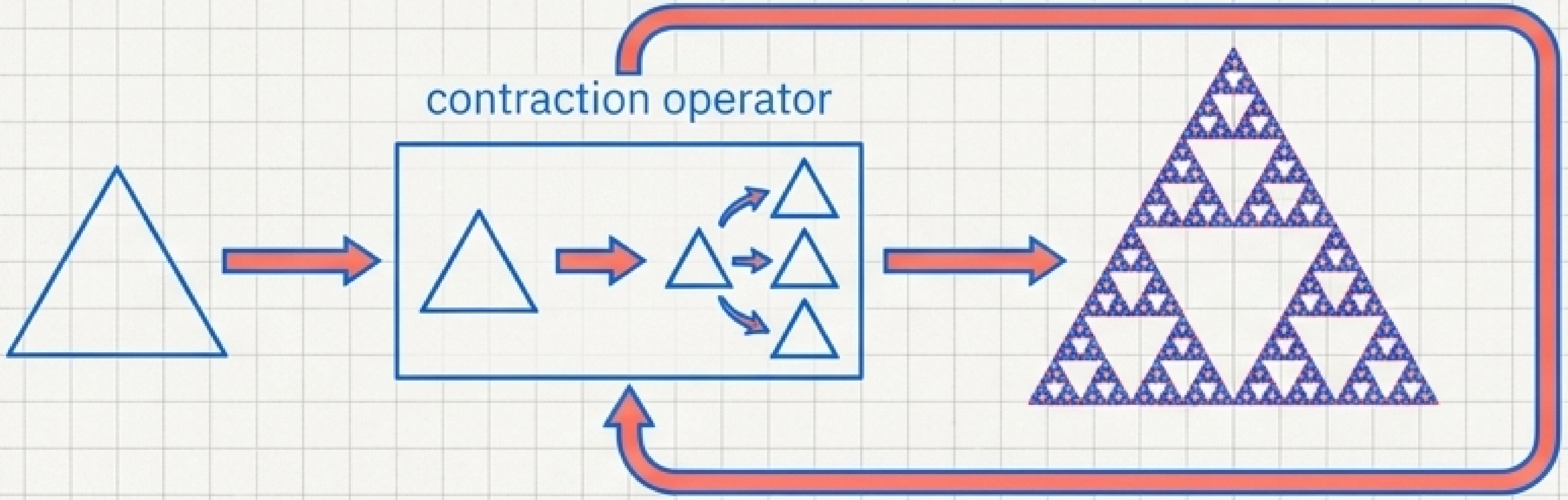
Expansion is accumulation. Contraction is generation.

Reachable Expansion (Lemma 1.1)

$$S_t \cup Adj(S_t) = S_{t+1}$$

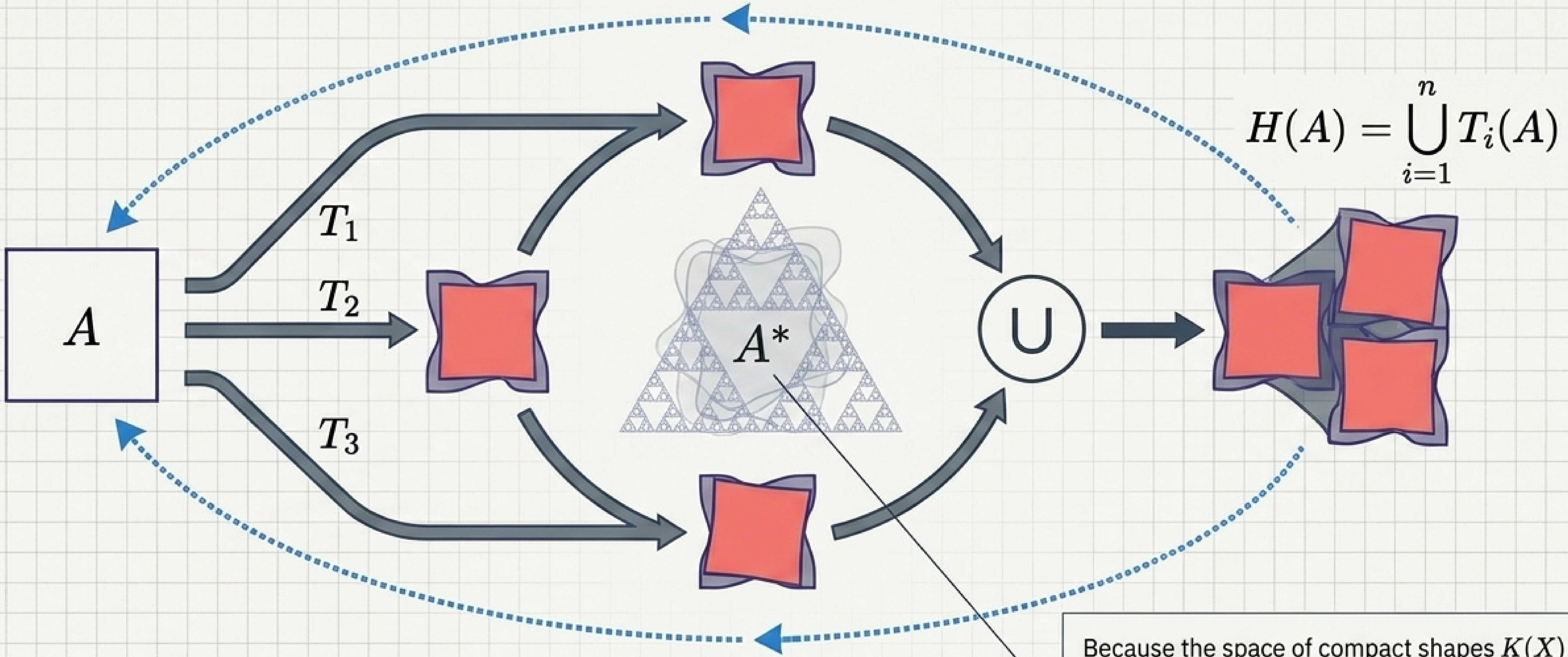


Hutchinson Operator (Definition 1.2)



Monotonic expansion grows without bound. It never converges; it is never self-similar.
Generation requires operators applied repeatedly to themselves to reproduce their own generating rule.

The Continuation Attractor Theorem

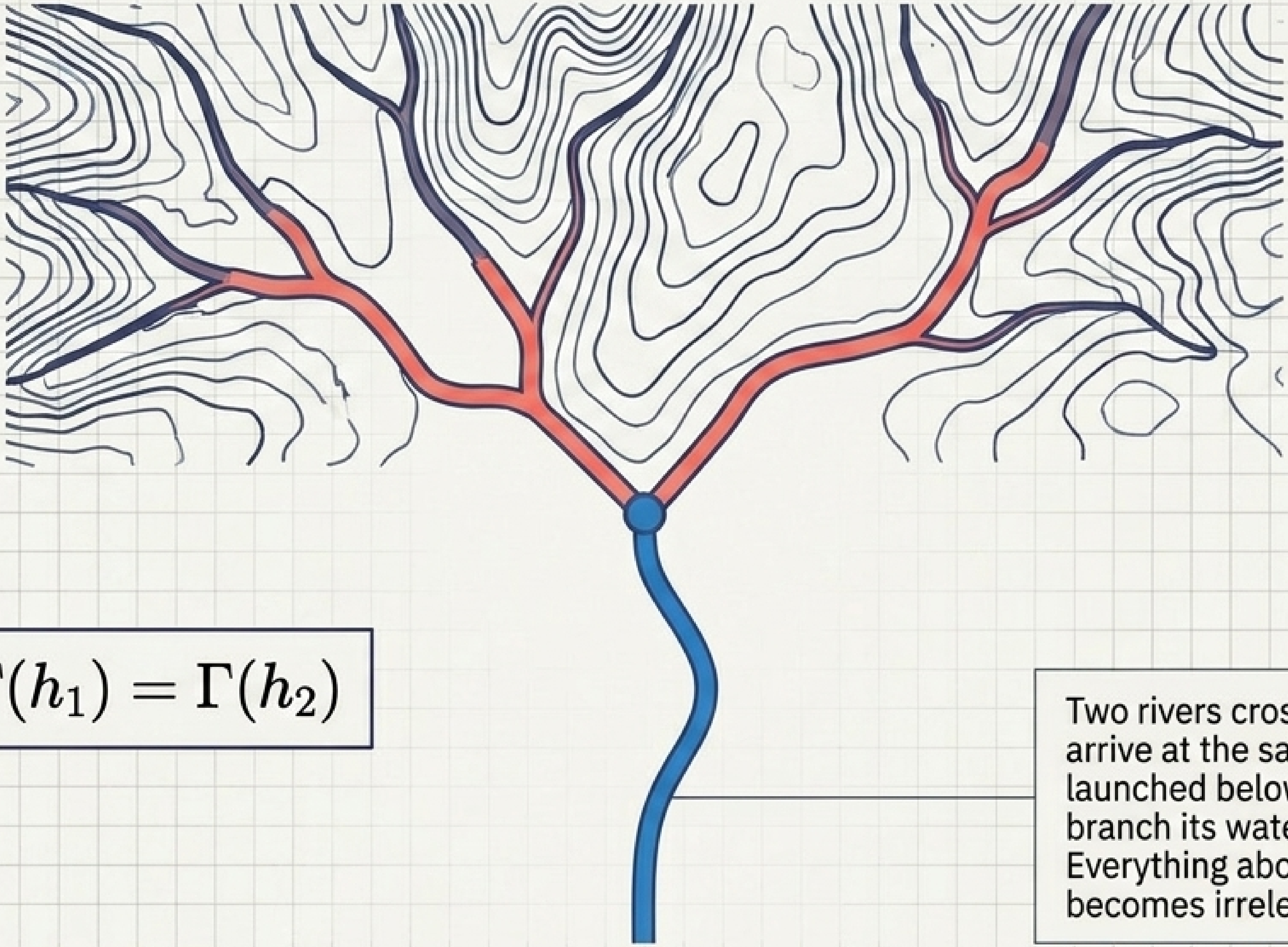


$$d_H(H^k(A_0), A^*) \le \frac{c^k}{1-c} d_H(A_0, H(A_0)) \to 0$$

Because the space of compact shapes $K(X)$ is complete under Hausdorff distance d_H, and H is a strict contraction ($c < 1$), the system guarantees a unique fixed point A^* where $H(A^*) = A^*$.

Histories merge when distinctions no longer alter the future.

Continuation Quotient

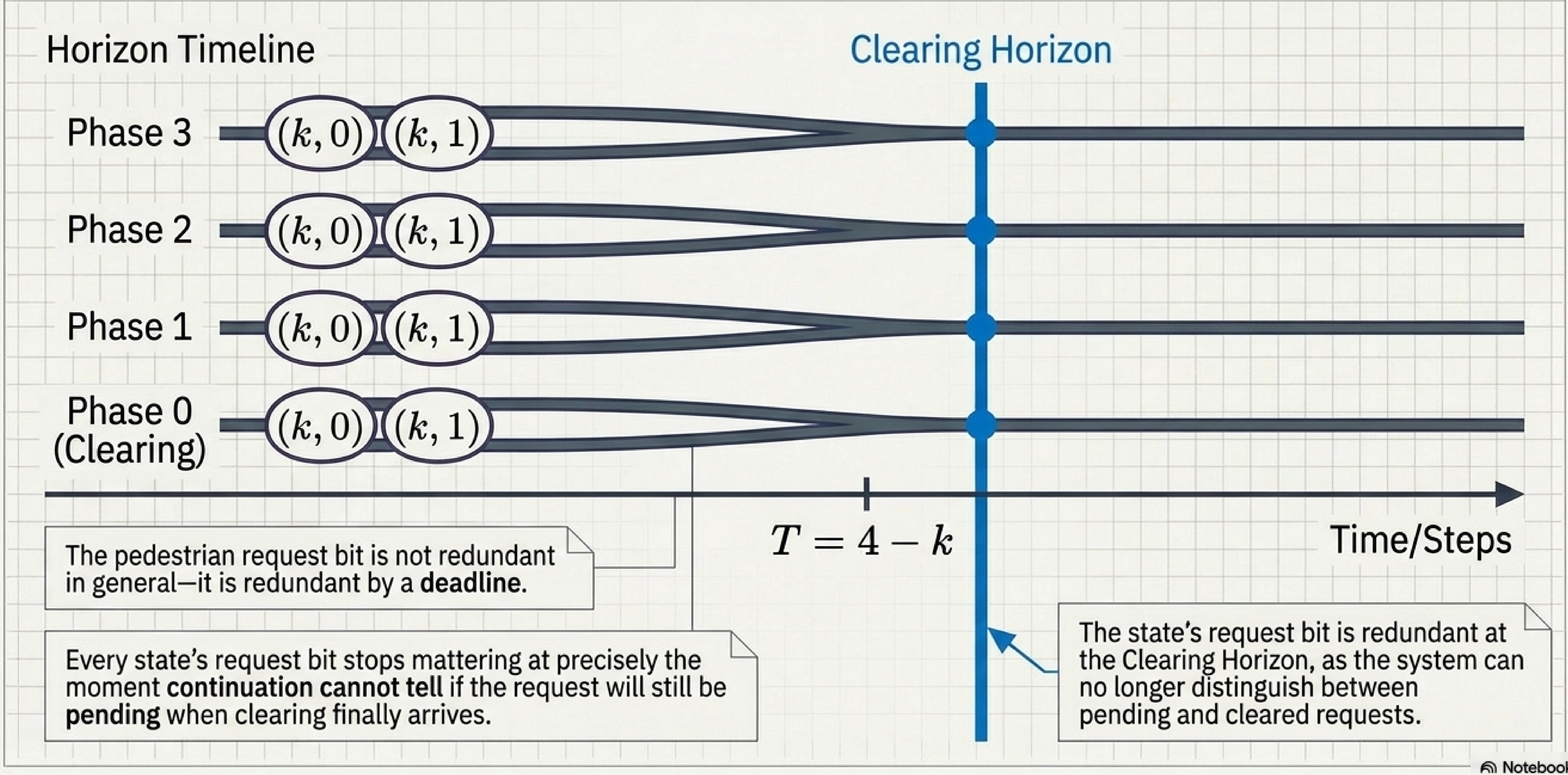


Definition

$$h_1 \sim_\Gamma h_2 \Leftrightarrow \Gamma(h_1) = \Gamma(h_2)$$

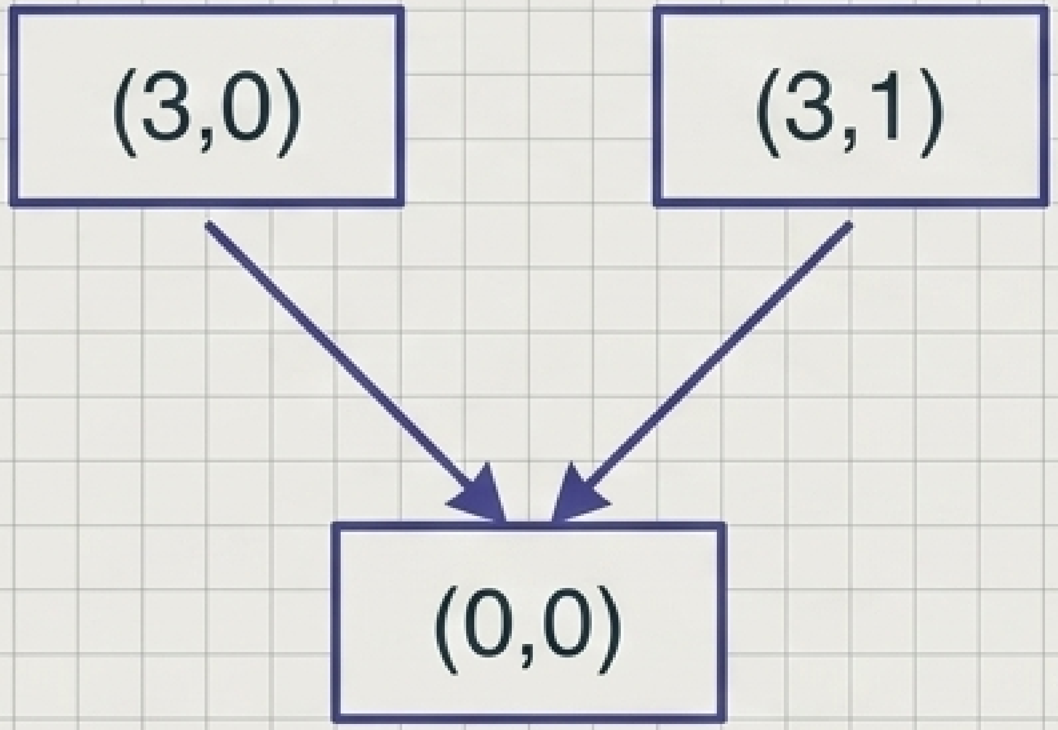
Two rivers crossing different terrain arrive at the same confluence. A boat launched below cannot tell which branch its water came from. Everything about their separate histories becomes irrelevant downstream.

Eventual periodicity bounds the check.



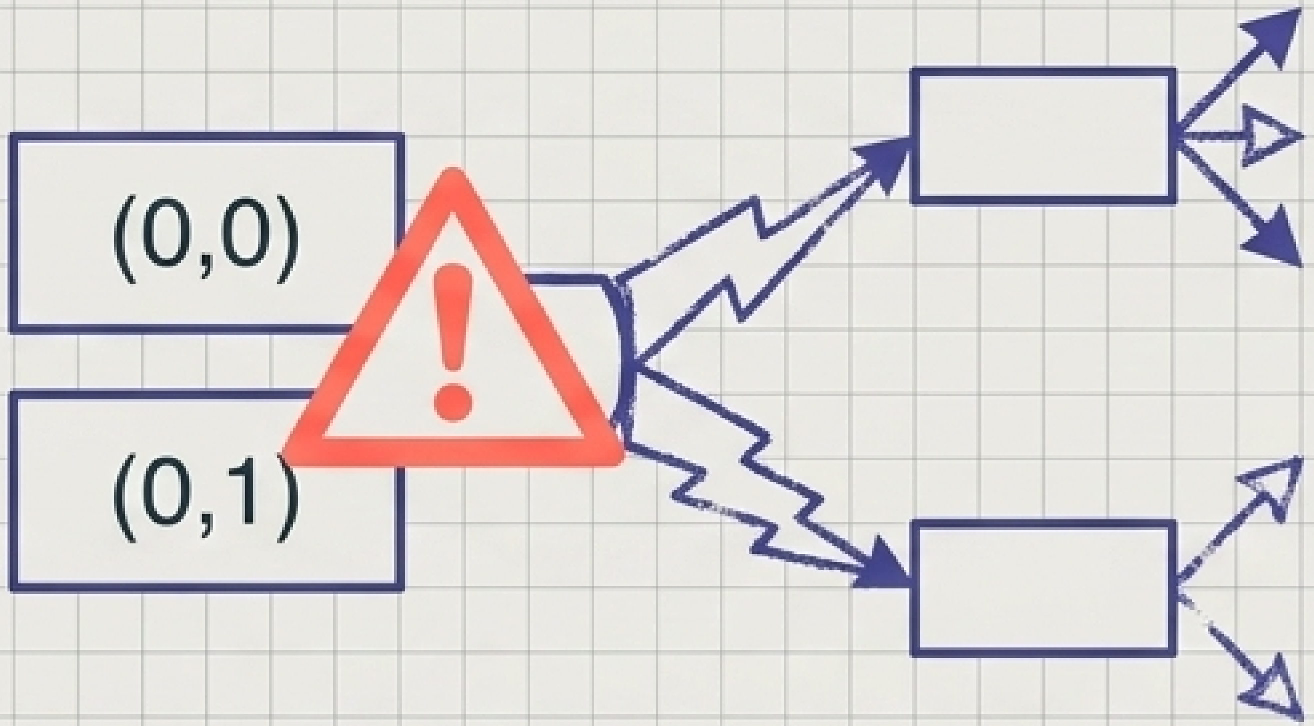
Aligning the mechanical quotient with the continuation quotient

Exact Design: Clear on Arrival at Phase 0



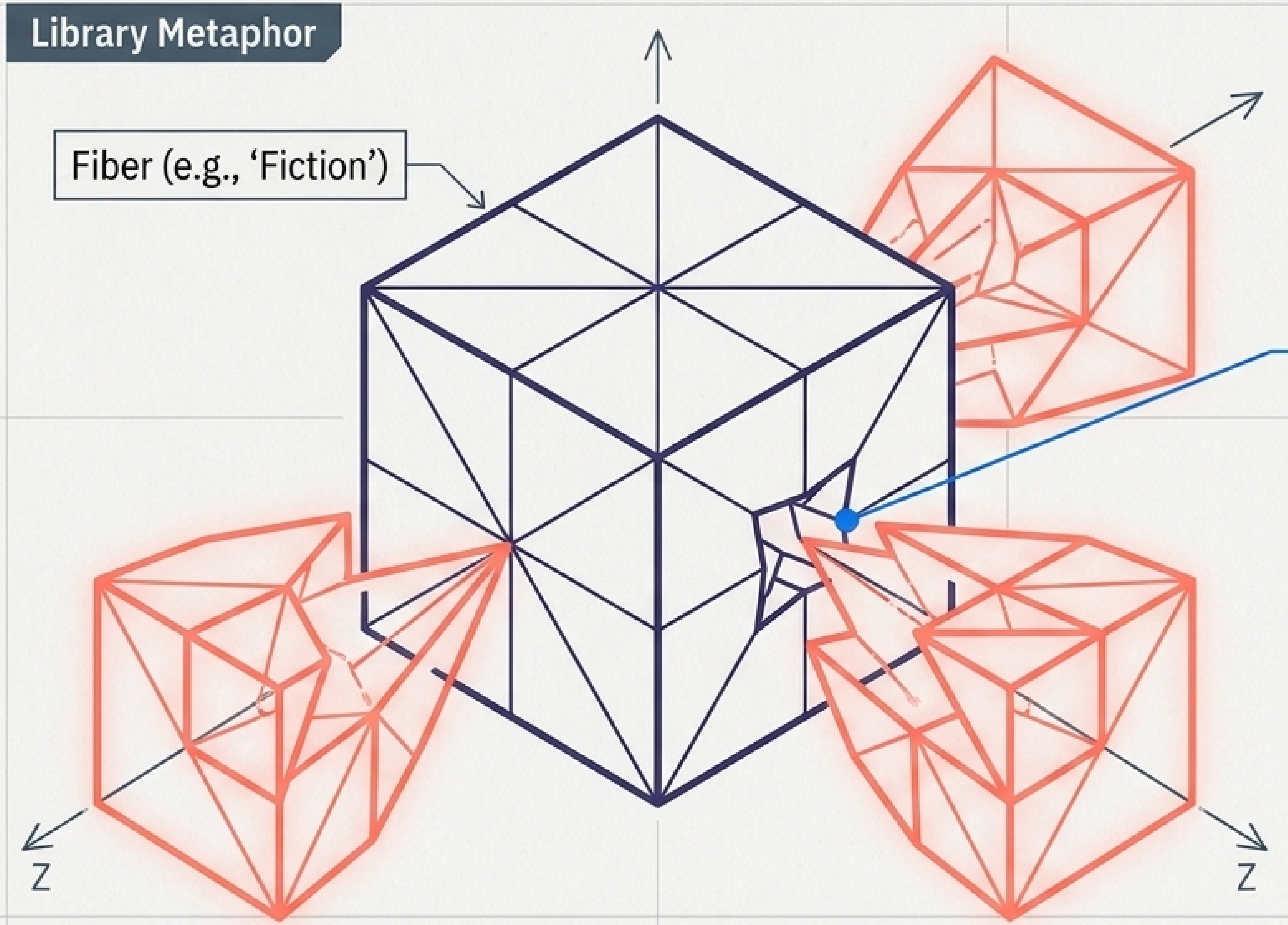
Identifies the two states about to make the transition. $R_1(3,0) = R_1(3,1) = \{(0,0)\}$.

Flawed Design: Clear While Sitting at Phase 0



Merges states with genuinely different one-step futures. A state at red with no request can branch; one with a request cannot.

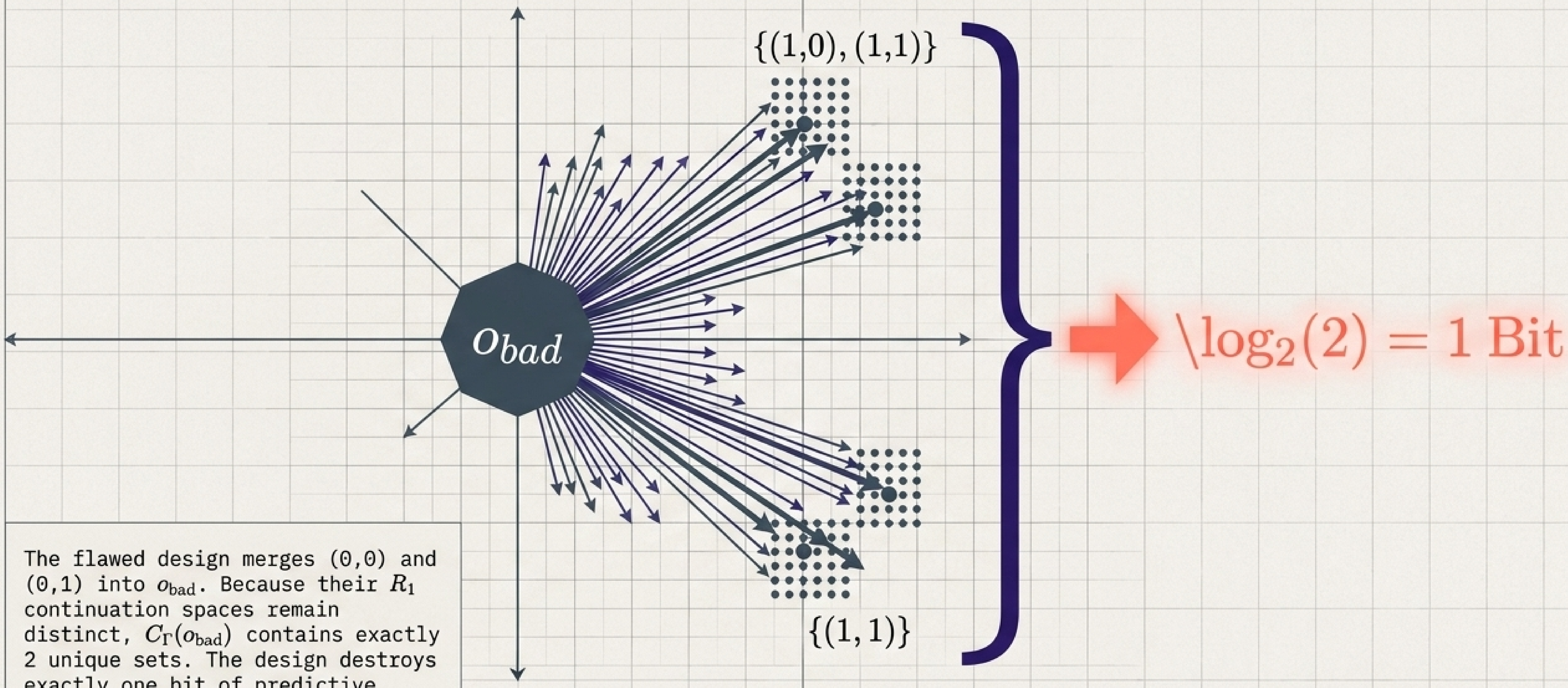
Measuring what is lost when distinctions are erased.



$$L(o) = \log|C_\Gamma(o)|$$

A zero-loss quotient (L=0) refines $\sim_\Gamma$. When a quotient is coarser, multiple distinct continuation spaces get trapped in the same fiber. The loss is the log of the number of unique futures an observer is suddenly blinded to.

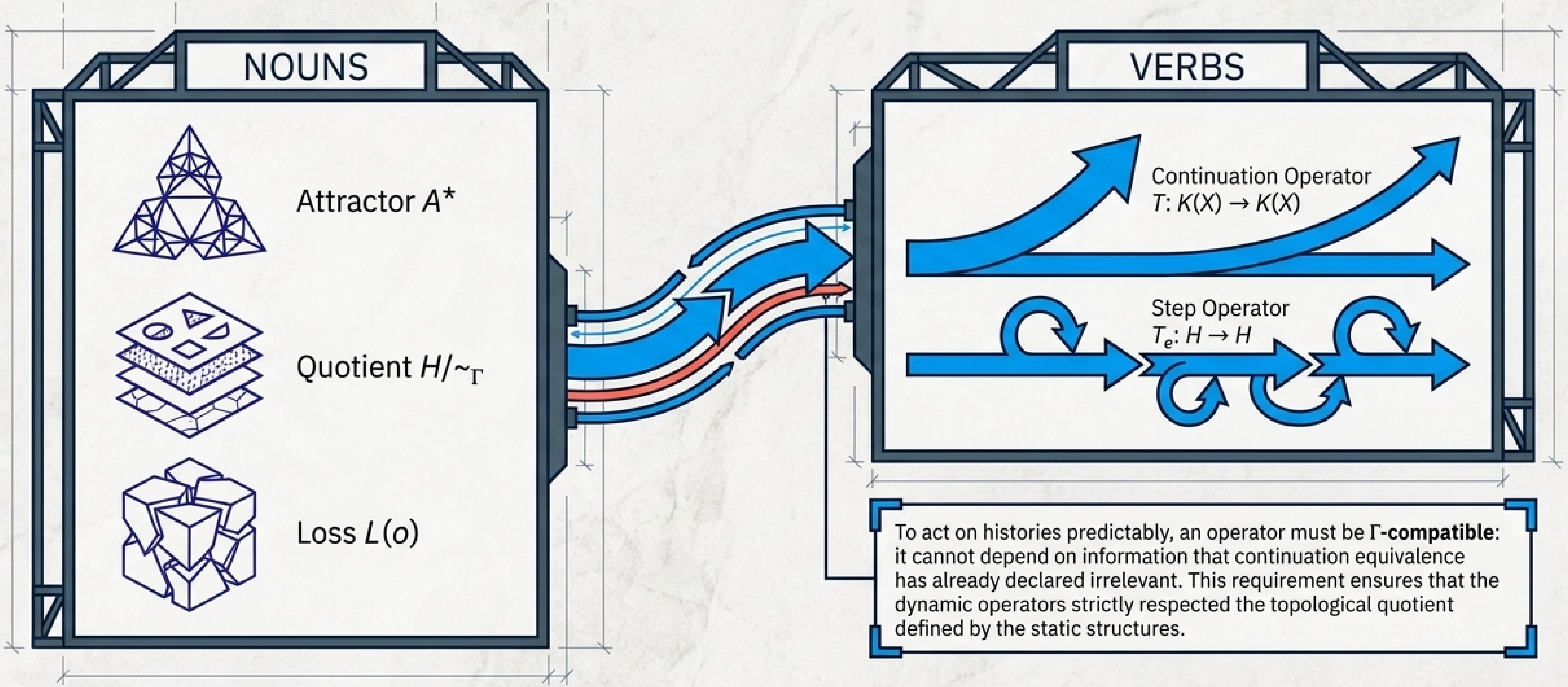
Pinpointing a single bit of mathematical loss.



The flawed design merges (0,0) and (0,1) into o_{bad}. Because their R_1 continuation spaces remain distinct, $C_\Gamma(o_{\text{bad}})$ contains exactly 2 unique sets. The design destroys exactly one bit of predictive power at this specific fiber.

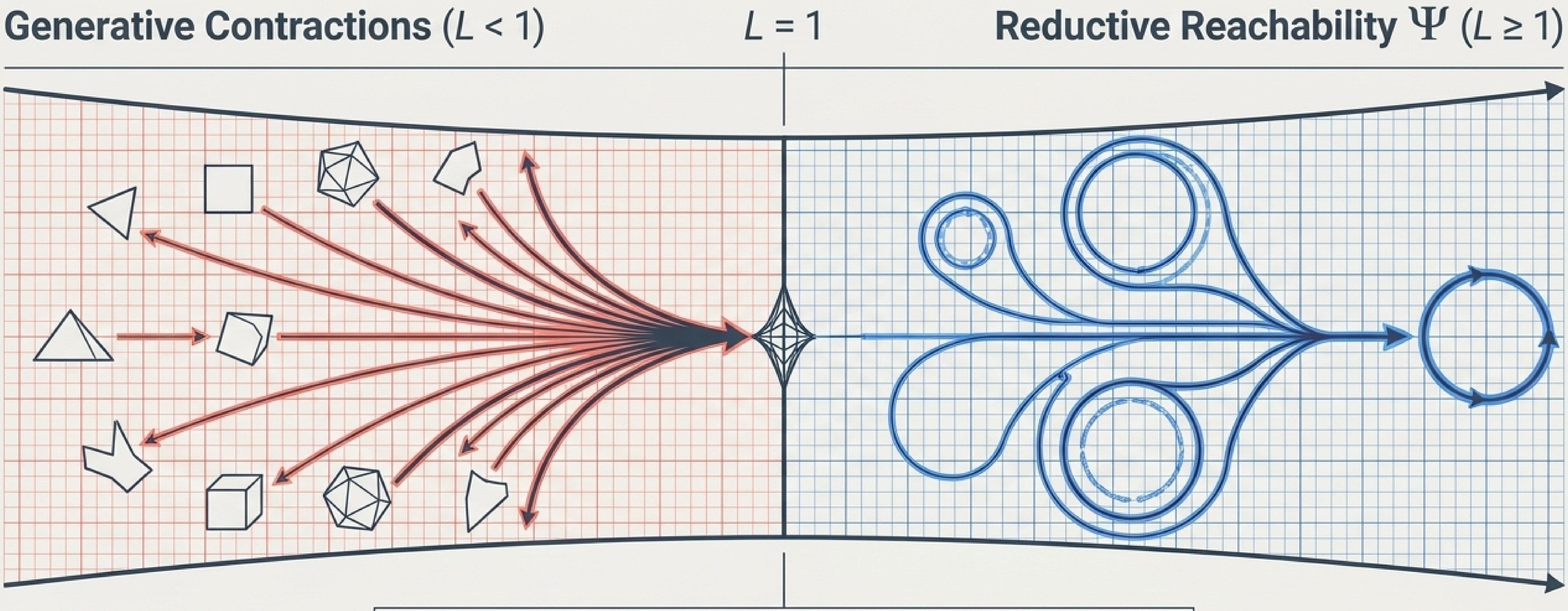
From Nouns to Verbs: The Continuation Operator

A **continuation operator** is a Lipschitz map on the space of shapes. It must satisfy $d_H(T(A), T(B)) \le L \cdot d_H(A,B)$.



To act on histories predictably, an operator must be **Γ-compatible**: it cannot depend on information that continuation equivalence has already declared irrelevant. This requirement ensures that the dynamic operators strictly respected the topological quotient defined by the static structures.

Generation and collapse are regions of a single parameter.



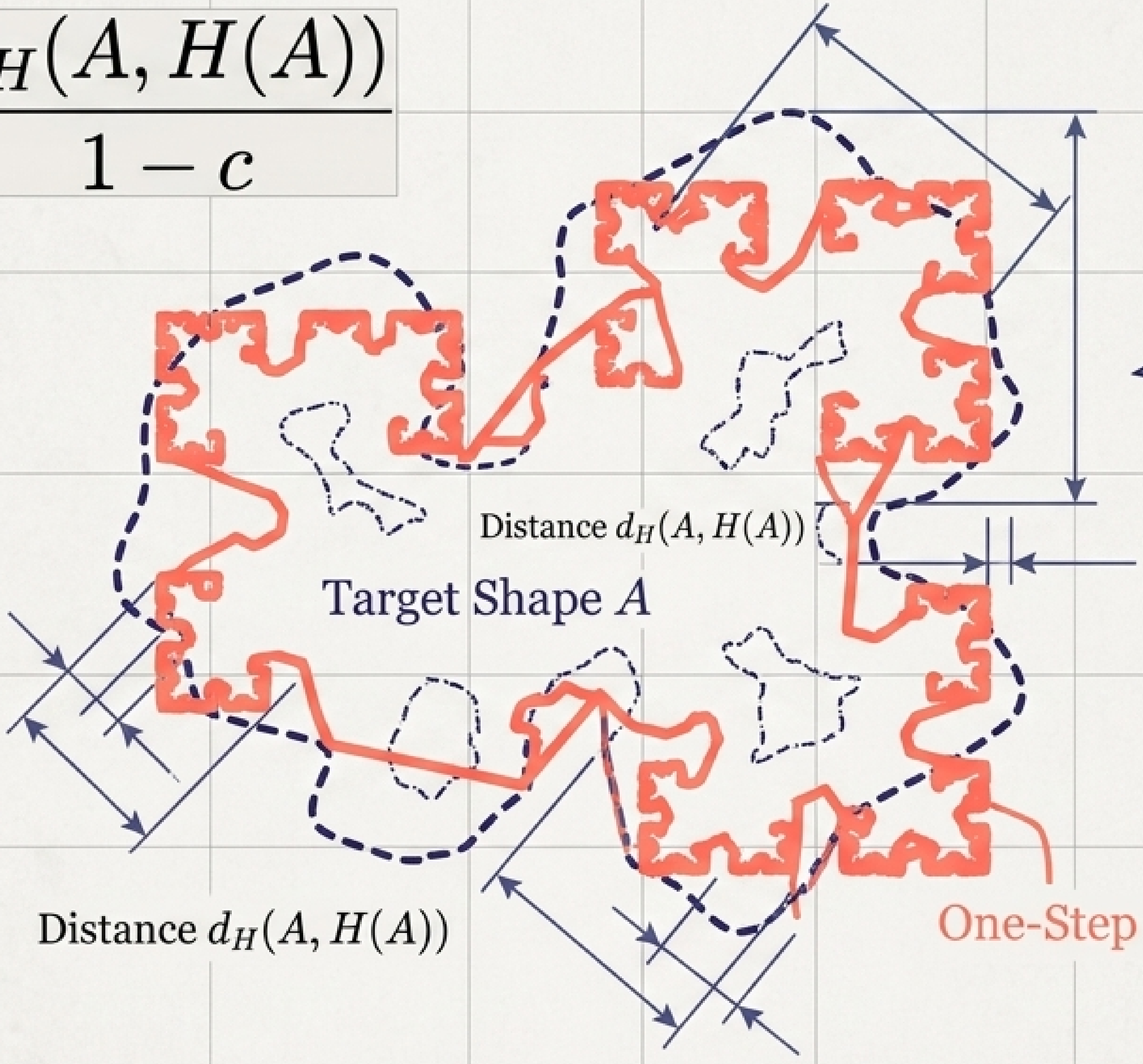
Building detail and erasing detail look like opposites. But both are instances of Continuation Operators. They differ only in one number: the Lipschitz constant. A contraction pulls shapes together; the reachability map Ψ advances them toward eventual periodicity.

The Operator Matrix

	Generative Contractions (Ch 1)	Reductive Reachability Ψ (Ch 2/3)
Mechanism	Builds self-similar attractors via fractional scaling	Advances finite-horizon reachable sets via union
Lipschitz Constant (L)	$L < 1$ (Strict contraction)	$L \geq 1$ (Non-contractive)
Effect on Distinctions	Pulls all shapes closer together	Merges equivalent histories exactly, eliminating dead data
Limit Behavior	Geometric convergence to a unique shape **A***	Eventual periodicity (bounded by $2 \cdot 2^{\lvert X \rvert}$)

The Collage Theorem: Bounding the infinite with one step.

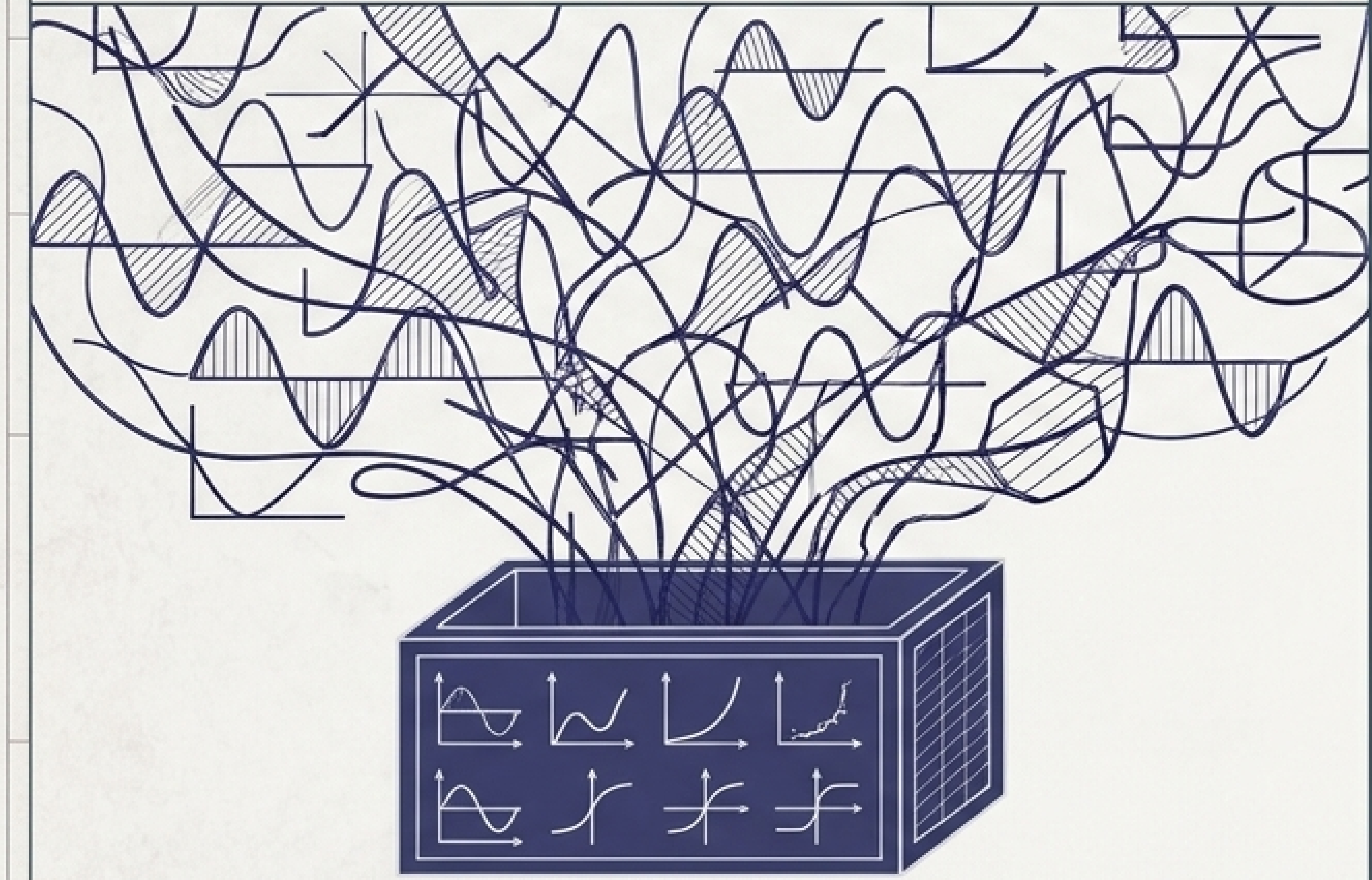
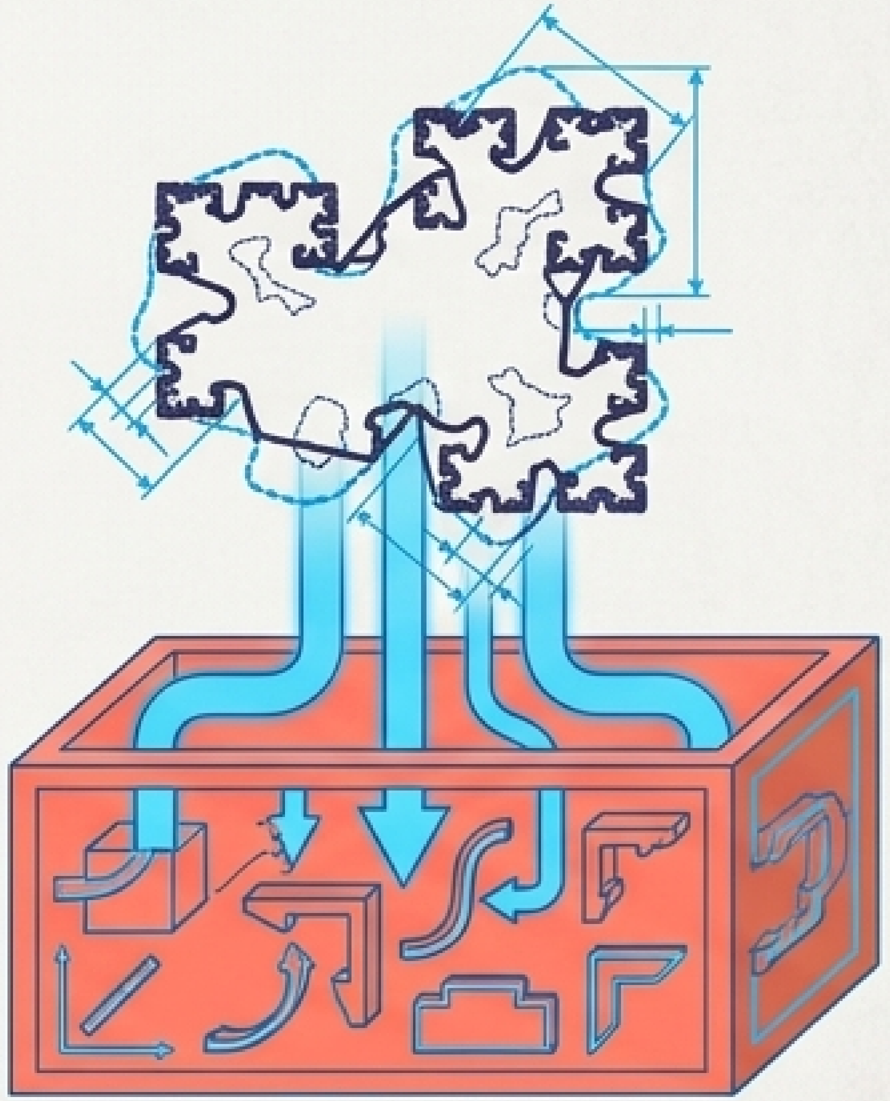
$$d_H(A, A^*) \leq \frac{d_H(A, H(A))}{1 - c}$$



To find an operator family whose attractor A^* is close to target A, we only need to find a one-step collage $H(A)$ close to A.

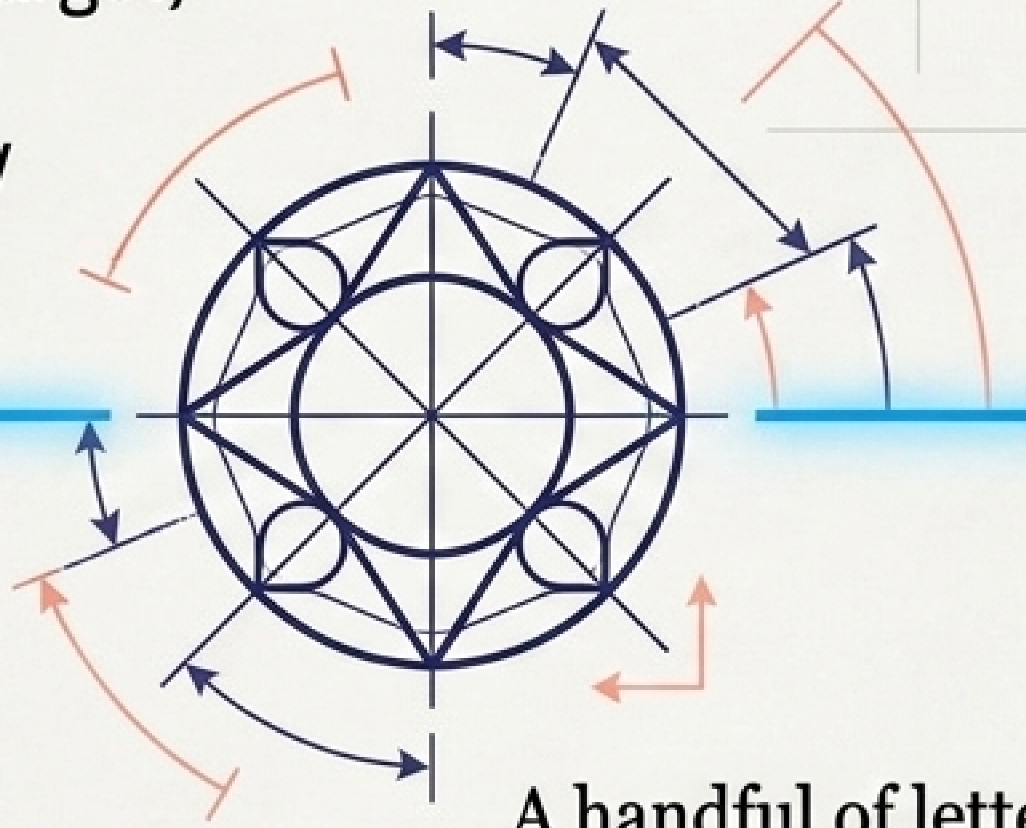
It converts an infinite limit problem into a single-step geometric fitting problem.

The Universality Gap.

Fixed Generators	**Target-Dependent Generators** (Theorem 5.2)
	
Classical theorems guarantee a fixed class of building blocks (e.g., polynomials, activation functions) can approximate any target.	Currently, every compact target in $\mathcal{K}(X)$ is approximable, but the generator family must be custom-built after the target and tolerance ε are known.

The Ultimate Open Question.

Universal Continuation Generation.
Does there exist a finite family of contractions, fixed once and independent of any target, capable of reaching arbitrary Hausdorff-distance accuracy for any admissible geometry?



A handful of letters generates every unwritten word.
A handful of rules generates every unplayed game.

Whether a single, fixed vocabulary of operators can generate all of continuation geometry remains the frontier.